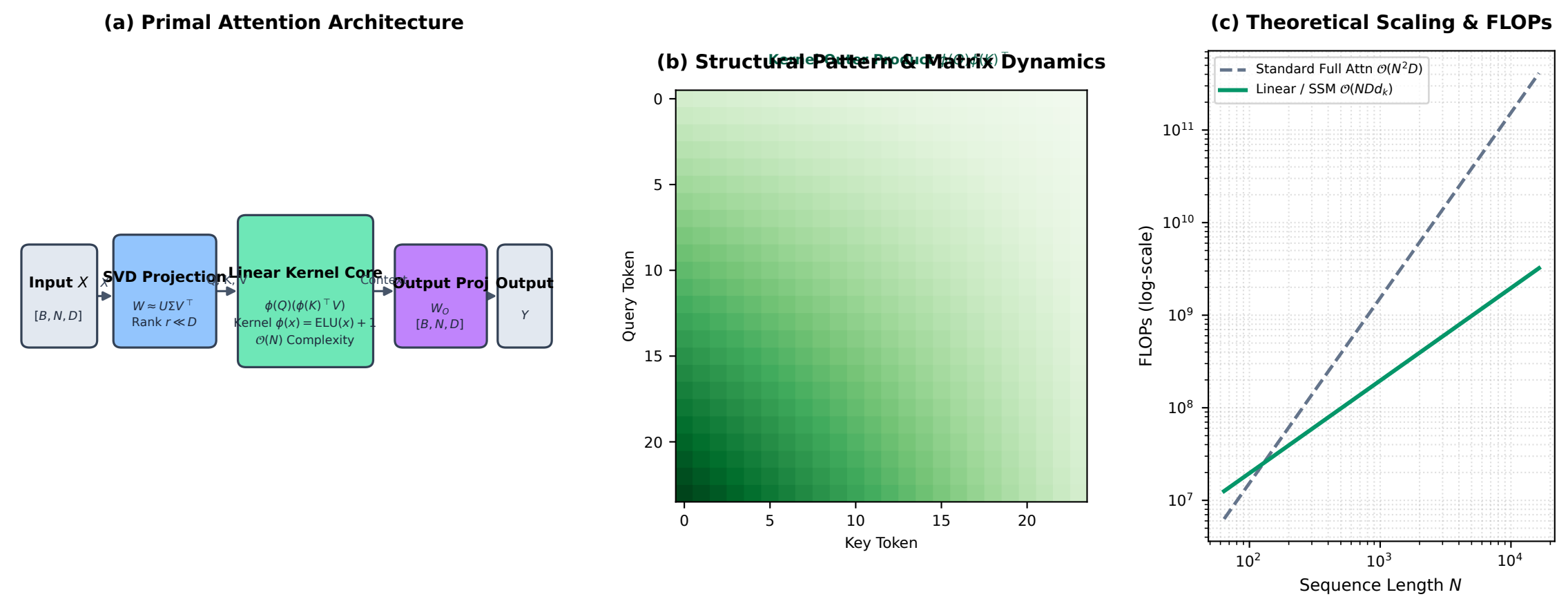
(a) Primal Attention Architecture
Input X
$[B, N, D]$
SVD Projection
$W \approx U\Sigma V^\top$
Rank $r \ll D$
Q, K, V
Linear Kernel Core
$\phi(Q)(\phi(K)^\top V)$
Kernel $\phi(x) = \mathrm{ELU}(x) + 1$
$\mathcal{O}(N)$ Complexity
Context
Output Proj
W_O
$[B, N, D]$
Output
Y
(b) Structural Pattern & Matrix Dynamics
Kernel Outer Product $\phi(Q)(\phi(K)^\top$
Query Token
Key Token
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Linear / SSM $\mathcal{O}(NDd_k)$
FLOPs (log-scale)
Sequence Length N
10^7
10^8
10^9
10^{10}
10^{11}
10^2
10^3
10^4